CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

May 2012

CL211 Environmental Engineering-I

Date: 18.05.2012, Friday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q-1 (a) Define per capita water demand. Describe various factors affecting per capita water demand. [05]

(b) Explain following terms (any one) [02]

(1) surface overflow rate (2) detention period (3) velocity gradient (4) Environmental Engineering

Q-2 (a) Draw a neat sketch of treatment-scheme used for treatment of water drawn from a perennial river. Also mention functions of each treatment unit. [06]

(b) Enlist salient points to be kept in mind for site selection of a water treatment plant. [04]

(c) Write short note (any one): [04]

(1) Clariflocculator (2) Prechlorination and Postchlorination (3) Stoke's law for settling velocity and its limitations.

OR

Q-2 (a) Discuss in details various mechanisms of chemical coagulation. [07]

(b) For a flocculator unit treating 300 m^3/h of water, determine power requirement and area of paddles. Assume detention time = 20 min., average value of $G = 40\ s^{-1}$, paddle tip velocity = 0.6 m/s and relative velocity = 0.75 of paddle tip velocity, dynamic viscosity = 1.0087×10^{-3} Pa.S, CD = 1.8 [04]

(c) Explain briefly design period. [03]

Q-3 (a) For a horizontal flow rectangular sedimentation basin, prove that settling velocity of a particle is independent of basin depth. Also write the assumptions made. [05]

(b) A settling tank is designed for an overflow rate of 3000 $lit/m^2/h$. What percentage of particles of diameter 0.025 mm (specific gravity: 2.65) will be removed at 20 $^{\circ}C$? [04]

(c) Define coagulation. Why coagulation is used in water treatment? Name a few common coagulants used and discuss one of them in detail. [05]

OR

Q-3 (a) What is sedimentation? Explain briefly different types of settling. [06]

(b) Design a rectangular sedimentation tank to treat 2 million liters of water per day of coagulated water. Assume detention period to be 3h and horizontal flow velocity = 75 mm/min and water depth of 3.0 m. Calculate the surface overflow rate and weir loading rate for this tank. [05]

(c) What will be the pH of a mixture formed by mixing following two solutions? Solution [03]

A: volume 300 mL, pH: 7; Solution B: volume 700 mL, pH 5.

SECTION - II

Q-4 (a) Enlist various types of intake structures and describe any one of them with neat sketch. [04]

(b) Chlorine usage in a treatment plant treating 25000 m^3/d of water is 9 kg/d. The residual chlorine after 10 min. contact is 0.2 mg/L. Calculate the chlorine dose in mg/L and chlorine demand. [03]

Q-5 (a) Design a bell mouth canal intake for a city of 100000 persons drawing water from a canal which runs for 10 hours in a day with a depth of 1.8m. Assume average water consumption = 150 LPCD. Assume velocity through screens and bell mouth to be less than 0.16 m/s and 0.32 m/s respectively. [06]

(b) Discuss various types of hardness normally found in water [04]

(c) Differentiate between a rapid sand filter and slow sand filter [04]

OR

Q-5 (a) Describe various actions that take place when water is filtered through a bed. [06]

(b) A filter unit is 4.5 m x 9.0 m in size. After filtering 10000 m^3/d of water, the filter is backwashed at a rate of 10 L/m^2/sec for 15 min. Calculate average filtration rate, quantity and percentage of water used in washing. Assume that filter is stopped for 30 min. everyday for backwashing. [04]

(c) Enlist various methods used for population forecast and briefly describe arithmetic increase method. [04]

Q-6 (a) Give logical reasons to prove that "indoor air pollution is much more dangerous than outdoor air pollution" [04]

(b) Draw a neat sketch of a house service connection and discuss functions of each component. [05]

(c) Enlist various types of traps used in drainage. Explain any one of them with neat sketch. [05]

OR

Q-6 (a) Discuss briefly the air conditioning system used for indoor ventilation. [06]

(b) Water is to be supplied to a town of 1.5 lakh population at the rate of 140 LPCD from a river, 1 km away. The difference in the elevation between the lowest water level in the intake and service reservoir is 20 m. Determine the size of main. If the pumps are to be operated for total of 8h/day and the efficiency of pump is 80%, determine hp of pump required. Assume friction factor as 0.03 and velocity of flow through main = 1.5 m/s. Assume maximum demand = 1.8 times average demand of water. [06]

(c) Draw a typical house drainage plan of a residential building [02]

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